

BHARATH SIMHA REDDY BHUREDDY

AI Backend Engineer | Real-Time Voice Pipelines & Production LLM Systems

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Summary

AI Backend Engineer specialized in building real-time voice pipelines, achieving **sub-800ms TTFA (p90)** and cutting LLM costs by **78%**. Built the **dvitva.ai** platform serving multiple clients on a serverless architecture, reducing server infrastructure costs by **30%**.

Skills

Programming	: Python, SQL
Voice AI & Audio	: WebSockets, Plivo Telephony, Silero VAD, Pipecat Smart Turn (ONNX), WebRTC AEC
GenAI & LLMs	: LangGraph, LangChain, Tool Calling, Structured Outputs, RAG , Streaming / SSE
Backend	: FastAPI, Pydantic, Motor (Async MongoDB), Redis, REST APIs, Webhooks, PostgreSQL, Supabase
Model Inference	: Hugging Face, ONNX Runtime, ctranslate2, IndicTrans2, NumPy
Cloud & DevOps	: RunPod (Serverless Containers), Docker, AWS ECS, Linux, Git, GitHub Actions (CI/CD)
Observability	: DeepEval, Locust (Load Testing), Pytest, LangSmith, Langfuse, Structured Logging

Experience

Dvitva AI (Child Company of Alonzo AI)

Hyderabad, India

Associate AIML Engineer

Apr 2026 – Sept 2026

- Architected **dvitva.ai** voice backend using **FastAPI** and **WebSockets**, streaming LLM chunks into TTS via **asyncio** and **Redis** to hit **sub-800ms TTFA (p90)**.
- Engineered serverless **ASR/TTS** microservices on **RunPod**, cutting model cold starts to **<200ms** via runtime memory tuning and warm container pools.
- Automated telephony provisioning via **Plivo** and enforced strict **Pydantic** validation schemas, eliminating mid-call tool execution schema failures.

Bodega Delivery

Guntur, AP, India

AI/ML Engineering Intern

Jul 2025 – Jan 2026

- Shipped a real-time **FastAPI** microservice using **Gemini Vision API** to verify food freshness and expiry labels, reducing delivery defects and complaints by **20%**.
- Built an enterprise RAG pipeline using **BM25 retrieval** across **60+ partner documents**, cutting lookup time by **50%**; containerized via **Docker** with **GitHub Actions** CI/CD.

Projects

Conversational Voice Agent Engine | *FastAPI, WebSockets, Plivo, RunPod, Redis, Silero VAD* | [Live](#)

2026

- Engineered bi-directional audio streaming infrastructure over WebSockets for PSTN telephony, hitting sub-800ms TTFA (p90) and sustaining low-latency concurrent call streaming.
- Integrated a two-tier VAD (Silero + Pipecat ONNX) for semantic turn boundary detection, eliminating barge-in false positives and preventing premature caller interruptions.

Indian Law Hybrid RAG Platform | *FastAPI, BM25, FAISS, Supabase, Hugging Face* | [GitHub](#)

2026

- Architected a hybrid retrieval pipeline pairing BM25 sparse lexical search with FAISS dense vector retrieval over legal statutes, improving citation retrieval recall by 35%.
- Deployed containerized inference service to Hugging Face Spaces with asynchronous query telemetry and relevance feedback logging to Supabase.

TradeMinds – Autonomous AI Paper Trading System | *FastAPI, GPT-4o-mini, Supabase, yfinance* | [GitHub](#)

2026

- Engineered an autonomous trading agent executing dynamic financial tool-calling via GPT-4o-mini and yfinance APIs to evaluate market indicators and execute paper trades.
- Implemented persistent trade journaling and ledger state management in Supabase, using idempotent execution checks to eliminate duplicate trade order mutations.

Education

Annamacharya Institute of Technology and Sciences

Rajampet, India

B.Tech, Artificial Intelligence and Machine Learning – **CGPA: 8.43/10**

2022 – 2026